

Divyanshu Raj

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EDUCATION

Acharya Institute of Technology

Bachelor of Engineering – Computer Science & Engineering | CGPA: 8.3/10

2023 – 2027

Bengaluru, India

Coursework: Data Structures & Algorithms, Computer Networks, Operating Systems, DBMS, OOP, Software Engineering

PROFESSIONAL EXPERIENCE

Pratinik Infotech

Machine Learning Intern

Feb 2026 – Apr 2026

Bengaluru, India

- Built a machine failure prediction model on the AI4I 2020 dataset (10,000 records, 6 sensor features) using Random Forest; achieved 98.4% accuracy and 95.4% ROC-AUC on a 2,000-record holdout set.
- Identified Torque (30.1%) and Rotational Speed (23.1%) as top failure predictors via feature importance analysis across 5 failure modes (TWF, HDF, PWF, OSF, RNF).
- Automated preprocessing and model testing pipelines in Python (Pandas, NumPy, scikit-learn), reducing manual data preparation effort across multiple datasets.
- Assisted senior engineers with model deployment workflows, gaining hands-on exposure to production ML system integration.

PROJECTS

AI Software Review Platform | Python, FastAPI, Gemini AI

- Built an AI-powered code review platform supporting individual Python files and ZIP-based projects, with automated multi-file analysis across entire codebases.
- Automated code analysis across 4 quality dimensions (readability, performance, security, maintainability) and 4 issue categories, with normalized 0–10 project health scoring.
- Integrated Google Gemini AI with structured JSON responses to deliver consistent reviews, strengths, weaknesses, suggestions, and quality metrics.
- Delivered an end-to-end workflow covering file upload, AI analysis, project aggregation, and a web-based dashboard for reviewing and comparing results.

Personal AI Assistant | Python, FastAPI, ChromaDB, Sentence Transformers

- Built a modular AI assistant integrating 6+ core subsystems – memory management, RAG, planning, tool execution, and unified context generation.
- Developed an end-to-end RAG pipeline with overlapping chunking (800-char chunks, 100-char overlap) and vector-based retrieval via ChromaDB; processed a 3,775-char document into 6 searchable chunks with metadata traceability.
- Designed a planner-executor architecture supporting 7 intent workflows: calculations, web search, job analysis, email drafting, approval handling, and AI conversation.

Predictive Maintenance ML System | Python, scikit-learn, Random Forest, Pandas

- Built a machine failure prediction system on the AI4I 2020 industrial dataset (10,000 records, 3.39% failure rate) using Random Forest, achieving 98.4% accuracy and 95.4% ROC-AUC across 5 failure mode categories.
- Engineered features from 6 sensor inputs; performed feature importance analysis identifying Torque and Rotational Speed as dominant failure drivers.

AI-Based Parental Control System | Python, Computer Vision, FastAPI

- Built a facial recognition system that automatically switches device access modes based on the detected user, replacing manual parental controls.
- Implemented multi-layer authentication combining face verification, OTP-based access control, and secure biometric data storage.
- Developed a FastAPI backend on Linux for authentication, activity logging, and uninstall protection; tested and closed bypass vulnerabilities.

TECHNICAL SKILLS

Languages

Python, C, C++, Java, JavaScript, SQL

AI & Machine Learning

Machine Learning, scikit-learn, Random Forest, Computer Vision, LLM APIs, Gemini AI, Qwen3, Ollama, RAG, ChromaDB, Sentence Transformers

Web & Backend

FastAPI, REST API, Node.js, React.js, HTML, CSS

Systems & Linux

Linux/Ubuntu, Bash scripting, process management, TCP/IP, networking fundamentals

Databases

MySQL, MongoDB, ChromaDB

Tools & Platforms

Git, GitHub, VS Code, Agile, OOP

ACHIEVEMENTS & CERTIFICATIONS

- Secured **Top 10%** finish (23rd of 200+ teams) at **Hackaphasia Hackathon 2025** – systems and AI-focused challenge.
- Built a functional prototype under a 24-hour constraint at **AmbITION Hackathon 2025**, demonstrating rapid systems design under pressure.
- Certificate of Internship** – Machine Learning, Pratinik Infotech (Id: **PNC2602324**) [\[View Certificate\]](#)